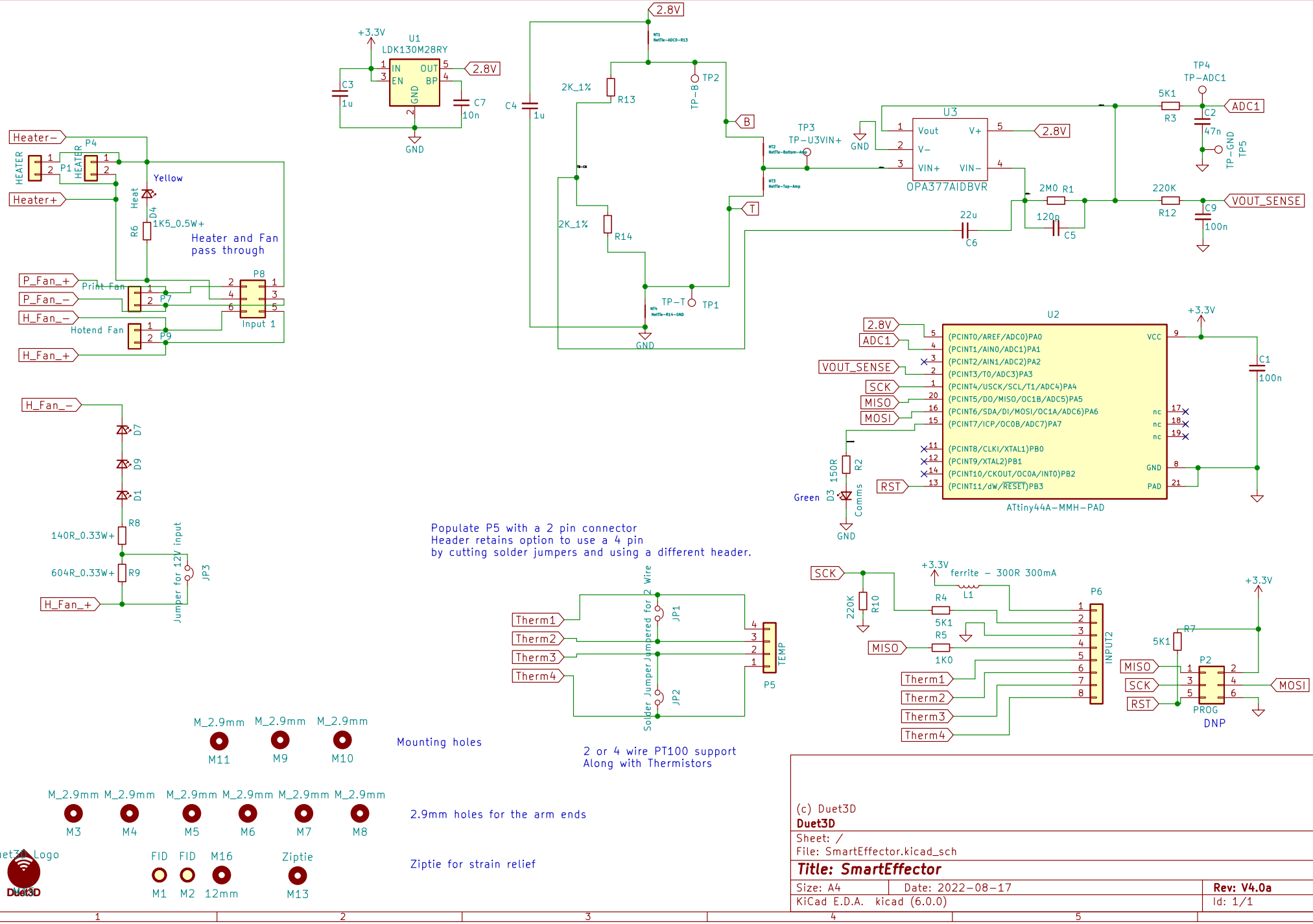

Heater−
Heater+
HEATER
P1
P4
Yellow
Heat
D4
1K5_0.5W+
R6
Heater and Fan
pass through
P_Fan_+
P_Fan_−
H_Fan_−
H_Fan_+
Print Fan
Hotend Fan
P7
P9
P8
Input 1
+3.3V
U1
LDK130M28RY
IN
EN
GND
OUT
BP
2.8V
C3
1u
C7
10n
C4
1u
2K_1%
R13
2K_1%
R14
TP−B
TP2
TP−T
TP1
B
T
TP3
TP−U3VIN+
U3
Vout
V−
VIN+
V+
VIN−
OPA377AIDBVR
2M0 R1
120n
C5
22u
C6
TP4
TP−ADC1
5K1
R3
C2
47n
TP−GND
TP5
220K
R12
C9
100n
ADC1
VOUT_SENSE
U2
(PCINT0/AREF/ADC0)PA0
(PCINT1/AIN0/ADC1)PA1
(PCINT2/AIN1/ADC2)PA2
(PCINT3/T0/ADC3)PA3
(PCINT4/USCK/SCL/T1/ADC4)PA4
(PCINT5/DO/MISO/OC1B/ADC5)PA5
(PCINT6/SDA/DI/MOSI/OC1A/ADC6)PA6
(PCINT7/ICP/OC0B/ADC7)PA7
(PCINT8/CLKI/XTAL1)PB0
(PCINT9/XTAL2)PB1
(PCINT10/CKOUT/OC0A/INT0)PB2
(PCINT11/dW/RESET)PB3
VCC
GND
PAD
nc
SCK
MISO
MOSI
RST
ATtiny44A−MMH−PAD
C1
100n
R2
150R
D3
Green
Comms
H_Fan_−
H_Fan_+
D7
D9
D1
R8
140R_0.33W+
R9
604R_0.33W+
Jumper for 12V input
JP3
Populate P5 with a 2 pin connector
Header retains option to use a 4 pin
by cutting solder jumpers and using a different header.
Therm1
Therm2
Therm3
Therm4
2 Wire
Jumpered for
Solder Jumper
JP1
JP2
TEMP
P5
2 or 4 wire PT100 support
Along with Thermistors
SCK
MISO
220K
R10
+3.3V
ferrite − 300R 300mA
L1
R4
5K1
R5
1K0
P6
INPUT2
R7
5K1
P2
PROG
DNP
MOSI
RST
M_2.9mm
M11
M9
M10
Mounting holes
M3
M4
M5
M6
M7
M8
2.9mm holes for the arm ends
Duet3D Logo
FID
M1
FID
M2
M16
12mm
Ziptie
M13
Ziptie for strain relief
(c) Duet3D
Duet3D
Sheet: /
File: SmartEffector.kicad_sch
Title: SmartEffector
Size: A4
Date: 2022−08−17
Rev: V4.0a
KiCad E.D.A. kicad (6.0.0)
Id: 1/1